

ELLIOT JERNG

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EDUCATION

Columbia University | MS in Neural Engineering

Sep 2026 - May 2028

Northeastern University | BS in Behavioral Neuroscience and Data Science

Sep 2022 - May 2026

- GPA: 3.93 / 4.0
 - summa cum laude honors
 - U.S. National Honor Society in Neuroscience
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EXPERIENCE

AbbVie | Cambridge, MA

Jul 2025 - Apr 2026

R&D Neuropsychiatry Co-op

- Developed target validation assays to measure neural activity and identify schizophrenia-relevant phenotypes.
- Built a Streamlit app for high-throughput spike-detection and customizable data analysis of calcium-imaging data.
- Computed quantitative features including spike count, intensity, area, frequency, and inter-spike interval.
- Deployed and presented the spike-detection and analysis software for team-wide use to optimize assay controls.
- Designed a glucose-uptake assay to quantify neural metabolism and activity for translational studies.
- Optimized positive controls to modulate glucose uptake bidirectionally, improving assay reproducibility.
- Applied siRNA-mediated knockdown of schizophrenia risk genes, revealing metabolic phenotypes in the assay.
- Established and maintained iPSC-derived neuronal cultures and clonal cell cultures.

Sabatini Lab | Harvard Medical School

Jul 2024 - Dec 2025

Research Assistant

- Conducted operant learning experiments integrating MATLAB, Arduino, SpikeGLX, fiber photometry, and optogenetics in a population of co-releasing neurons to study experience-dependent synaptic sign switching.
- Developed and presented bio-inspired artificial neural networks (ANNs) implementing variations of Dale's Law.
- Modeled synaptic plasticity in ANNs using a target-shifting task to induce adaptive weight changes.
- Sequentially reduced the number of updatable weights to test synaptic plasticity performance limits.
- Designed a custom reinforcement learning environment to model the operant conditioning behavioral task.
- Implemented PID-regulated temporal difference error to model dopamine modulation in the co-releasing circuit.
- Evaluated the role of co-release in ANNs by analyzing synaptic plasticity and adaptive learning performances.

Center for Translational Neuroimaging | Northeastern University

Jan 2024 - Jul 2024

Research Assistant

- Preprocessed fMRI data and aligned it to a standardized brain atlas.
 - Conducted rodent experiments investigating the effects of psilocybin following repetitive head injury.
 - Performed behavioral testing (rota-rod, beam walking) and MRI acclimation procedures.
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PUBLICATIONS

Li S, Wang W, Knipe G, Jerng E, Capelli P, Zhou C, Bilsel E, Sabatini BL. *Synaptic sign switching mediates online dopamine updates.* bioRxiv. 2025. [Preprint].

SKILLS

Programming & Tools: Python, NumPy, SciPy, pandas, MATLAB, SpikeGLX, R, Arduino, Streamlit

Machine Learning: Deep learning, spike detection, reinforcement learning, PyTorch, TensorFlow

Laboratory: Optogenetics, fiber photometry, iPSC culture, FLIPR, IHC, fluorescence imaging, perfusions, slicing

Languages: Chinese (Proficient), French (Proficient), English (Fluent)

Certificates: Youth Mental Health Ambassador at Star Vista HAPY

COURSEWORK

Deep Learning in Biomedical Signal Processing, Stat Analysis in Neural Data, Comp. Neuro., Linear Algebra, ML, AI

PROJECTS

Terminal-Bench Science (Ongoing work, 2026): [biophysical parameter fitting in single-neuron modeling](#)

- Contributing a task on biophysical parameter recovery in single-neuron modeling to Terminal Bench Science — a benchmark evaluating AI agents on real computational workflows from scientific research. This benchmark aims to drive scientific progress in AI agents at frontier labs by defining what those labs optimize for.

Classifying Hand Movements From sEMG Data: <https://github.com/elliottjerng/sEMG>

- Recreated results of "Characterization of a Benchmark Database for Myoelectric Movement Classification" by M. Atzori et al..
 - Built linear and non-linear classifiers from scratch, including LDA, SVM, KNN, and MLP.
 - Achieved equal/higher performance on all results with time domain features.
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VOLUNTEER

Sociedad Latina / Boston, MA

Aug 2022 - Dec 2022

Youth Arts Mastery Coordinator

- Developed and taught curriculum in music theory, ensemble play, and original composition.
- Created inclusive, encouraging classroom spaces for Latina youth.
- Instruct students in guitar, bass, drums, keyboard, and voice.

Star Vista Health Ambassador Program / Online

Oct 2021 - Dec 2021

Youth Mental Health Awareness Ambassador

- Trained to recognize and deal with various mental illnesses and neuropsychiatric disorders.
 - Presented mental health awareness slideshows to various groups of teenagers.
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